

NMAM INSTITUTE OF TECHNOLOGY, NITTE
 (An Autonomous Institution affiliated to VTU, Belagavi)
Seventh Semester B.E. (E&C) (Credit System) Degree Examinations
 January 2023

19EC701 - POWER ELECTRONICS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **each Unit**.*

Unit - I

1. a) Explain the control characteristics of following power devices.

- i. Thyristor
 ii. BJT

Marks BT* CO* PO*

10 L*2 1 1

- b) The waveforms of the transistor switch are as shown in Figure Q.1.b. The parameters are; $V_{CC} = 250$ V, $V_{BE(sat)} = 3$ V, $I_B = 8$ A, $V_{CE(sat)} = 2$ V, $I_{CE} = 100$ A, $t_d = 0.5$ μ s, $t_r = 1$ μ s, $t_s = 5$ μ s, $t_f = 3$ μ s, $f_s = 10$ kHz, and the collector-to-emitter leakage current is $I_{CEO} = 3$ mA. If the duty cycle is $k = 50\%$, calculate the power loss due to I_C during;

- i) turn-on $t_{on} = t_d + t_r$ ii) conduction period t_n

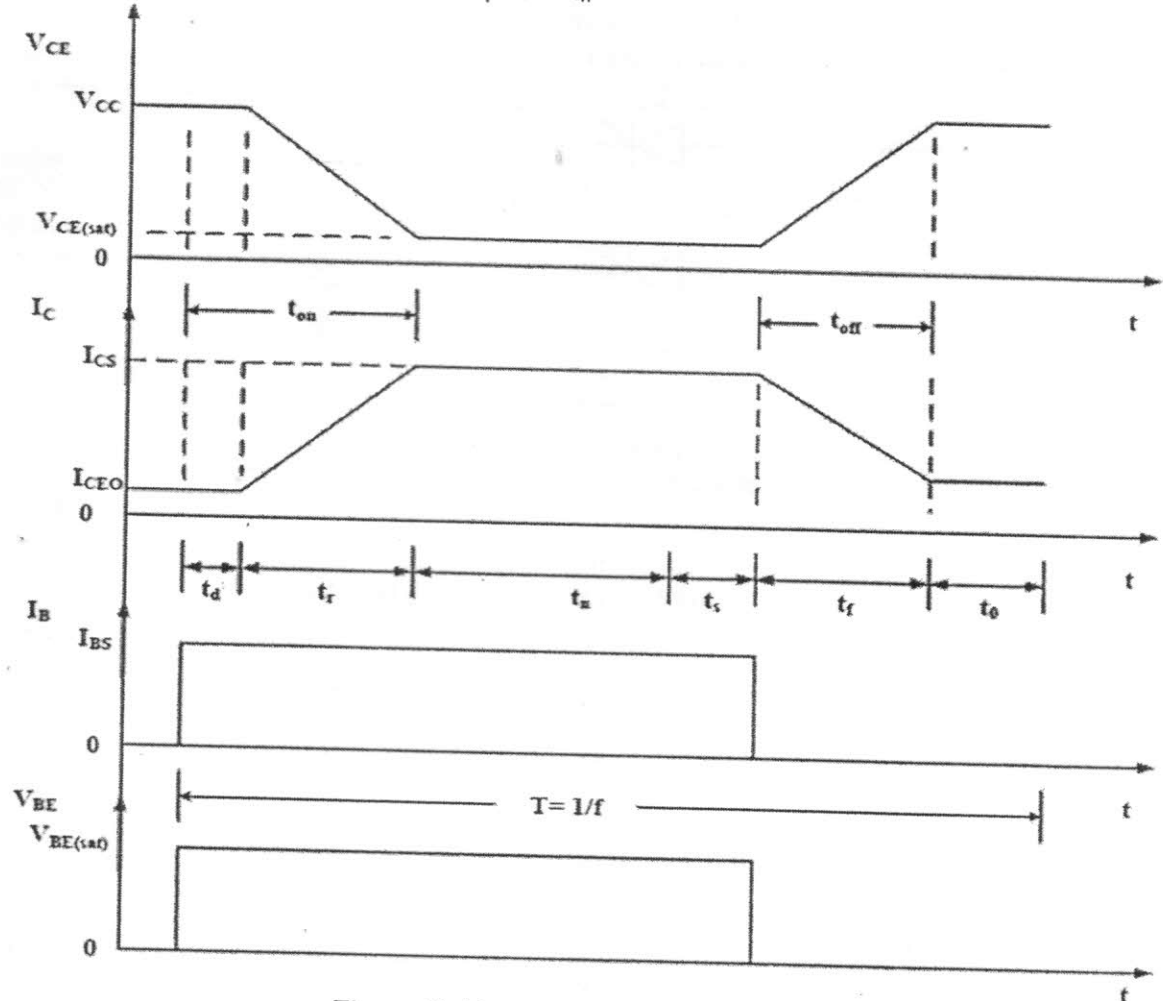


Figure Q.1.b.

10 L3 1 1

2. a) Explain the turn-on base drive control technique for a BJT.
 b) With neat waveforms, explain the switching characteristics of a Power MOSFET.

8 L2 1 1

7 L2 1 1

- c) The collector clamping circuit has $V_{CC} = 200 \text{ V}$, $V_{d1} = 2.5 \text{ V}$, $V_{d2} = 1.5 \text{ V}$, $V_{BE} = 0.7 \text{ V}$, $V_B = 20 \text{ V}$, $R_B = 3 \Omega$, $R_C = 2 \Omega$, and $\beta = 20$. Calculate the
- collector current without clamping.
 - collector-emitter clamping voltage, V_{CE} .
 - collector current with clamping.

5 L3 1 1

Unit – II

- With suitable mathematical analysis, prove that a thyristor is a latching device.
 - A thyristor is connected between a DC source of 150 V and a load. If the latching current of the thyristor is 80 mA , calculate the minimum width of the gate pulse that is required to turn-on the thyristor when the load
 - is purely inductive with an inductance of 150 mH .
 - consists of a resistor of 10Ω and inductor of 150 mH .
 - A thyristor has a current of 250 A while dv/dt capability is $70 \mu\text{s}$. If the circuit inductance is 0.2 mH , design a snubber circuit.
- Explain the dynamic turn-off characteristics for a thyristor.
 - An impulse commutated thyristor circuit is given in Figure Q.4.b. Determine the circuit turn-off time if $V_s = 230 \text{ V}$, $R = 20 \Omega$, $C = 10 \mu\text{F}$, and $V_0 = V_s$. Also, determine the voltage across the capacitor at the instant of commutation.

10 L2 2 1

6 L3 2 1

4 L3 2 1

8 L2 2 1

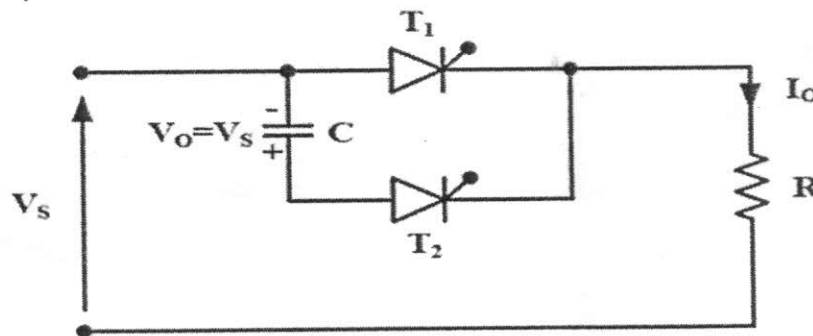


Figure Q.4.b

4 L3 2 1

- c) With necessary diagrams and waveforms, explain the resonant pulse commutation circuit.

8 L2 2 1

Unit – III

- If the converter shown in Figure Q.5.a has a purely resistive load of R and the delay angle is $\alpha = \pi/4$, determine a) rectification efficiency, b) form factor, FF, c) ripple factor, RF, d) transformer utilization factor, TUF, and e) peak inverse voltage, PIV, of the thyristor T_1 .

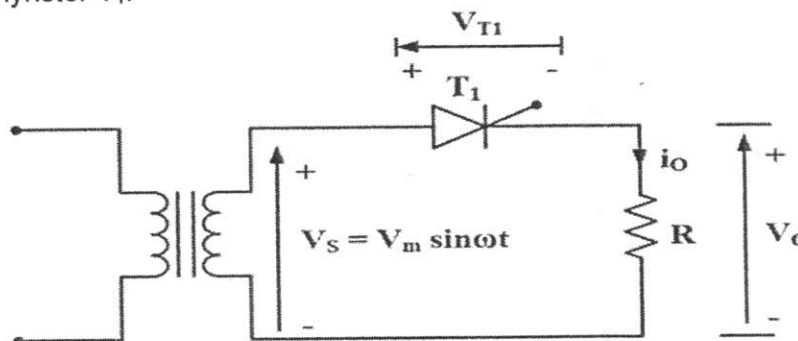


Figure Q.5.a

10 L3 3 1

- b) With neat diagrams and waveforms, explain the operation of single-phase full converter. Derive expressions for average and rms values of output voltage.
6. a) Explain the principle of phase control of an ac voltage controller using a single-phase unidirectional controller. Derive expressions for average and rms values of output voltage.
- b) A single-phase ac voltage controller as shown in Figure Q.6.b is controlled by a input supply voltage with $V_s = 230$ V, 60 Hz. Assume a purely resistive load of $R = 20 \Omega$. The firing angles of thyristors T_1 and T_2 are equal i.e. $\alpha_1 = \alpha_2 = \alpha = \pi/3$. Calculate i) the rms output voltage, V_o , ii) the input power factor, PF, iii) the average current of thyristors, and iv) rms currents of thyristors.

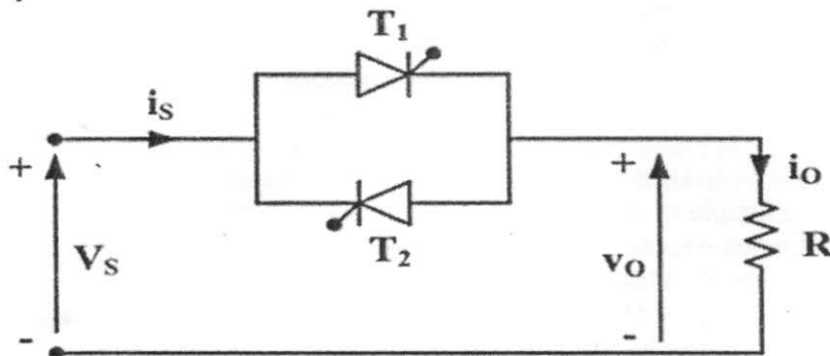


Figure Q.6.b

- c) A voltage source $v_s = 200 \sin(400t)$ supplies a resistive load of 200Ω through a pair of back-to-back connected thyristors. Calculate the average power in the load if the delay angles of the thyristors is $\pi/3$.

Unit – IV

7. a) A step-down chopper as shown in Figure Q.7.a is fed by a 230 V voltage source. The load is purely resistive with a value of 20Ω . Consider a practical chopper switch is with a voltage drop of $v_{ch} = 1.5$ V when it is on. The chopper is operated with a duty cycle of 60 %. Calculate
- the average output voltage, V_a ,
 - the rms output voltage, V_o ,
 - the input power, output power, and chopper efficiency, and
 - the effective input resistive of the chopper, R_i .

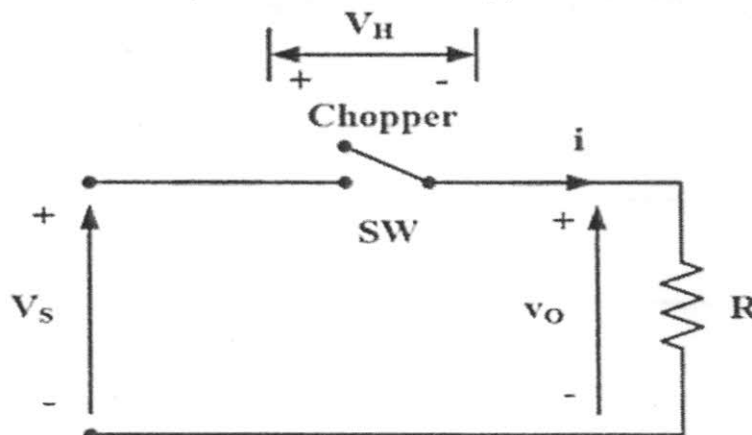


Figure Q.7.a

10 L2 3 1

10 L2 3 1

6 L3 3 1

4 L3 3 1

8 L3 4 1

- b) With neat circuit diagram and waveforms, explain the principle of operation of step-up chopper. Derive expression for instantaneous output voltage. 8 L2 4 1
- c) Input to the step-up chopper is 200 V. The output required is 600 V. If the conducting time of thyristor is 200 μ s. Compute:
i) Chopping frequency, ii) If the pulse width is halved for constant frequency of operation, find the new output voltage. 4 L3 4 1
8. a) Explain the basic elements of switching mode regulators. 6 L2 4 1
- b) The buck regulator has an input voltage of $V_S = 12$ V. The required average output voltage is $V_a = 5$ V at $R = 500 \Omega$ and the peak-to-peak output ripple voltage is 20 mV. The switching frequency is 25 kHz. If the peak-to-peak ripple current of inductor is limited to 0.8 A, determine (i) the duty cycle k , (ii) the filter inductance L , (iii) the filter capacitor C , and (iv) the critical values of L and C . 5 L3 4 1
- c) With neat circuit diagrams and waveforms, explain the working of a boost regulator. 9 L2 4 1
- Unit – V**
9. a) Explain the working principle of a single-phase full-bridge inverter with inductive load. Derive expression for rms output voltage. 8 L2 5 1
- b) A single-phase full-bridge inverter has a resistive load of $R = 5 \Omega$ and the dc input voltage is $V_S = 50$ V. Determine,
i) the rms output voltage at the fundamental frequency, V_1 ,
ii) the output power, P_o ,
iii) the average and peak currents of each transistor,
iv) the peak reverse blocking voltage V_{BR} of each transistor,
v) the total harmonic distortion, THD. 8 L3 5 1
- c) Briefly explain the following performance parameters of inverters.
i) Total harmonic distortion.
ii) Distortion factor. 4 L2 5 1
10. a) The single-phase half-bridge inverter has a resistive load of $R = 5 \Omega$ and the dc input voltage is $V_S = 50$ V. Determine:
(i) the rms output voltage at the fundamental frequency V_{o1} ,
(ii) the output power P_o ,
(iii) the average and peak currents of each transistor. 6 L3 5 1
- b) Explain multiple pulse-width modulation control technique. 6 L2 5 1
- c) Explain the working of transistor based current source inverter. 8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
